Sam Gijsen

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Machine learning researcher building foundation models and multimodal representation learning systems for physiological time series. Conditional generative modeling for out-of-distribution contexts. First-author publications at ICLR and ICML.

Selected Publications

Brain-Semantoks: Learning Semantic Tokens of Brain Dynamics with a Self-Distilled Foundation Model [link]

ICLR 2026, Gijsen, S, Schulz, M, Ritter, K

Self-distilled foundation model for brain dynamics. Pretrains in 2 hours, achieves strong downstream performance without finetuning, and exhibits log-linear scaling on cross-dataset downstream tasks.

Flow Matching with In-Context Priors for Out-of-Distribution Brain Dynamics [link]

Under Review; arXiv, Gijsen, S, Lukomski, M, Schulz, M, Ritter, K

Conditioning mechanisms for diffusion transformers enabling generation under unseen experimental conditions, with strong OOD generalization.

EEG-Language Pretraining for Highly Label-Efficient Clinical Phenotyping [link]

ICML 2025, Gijsen, S, Ritter, K

First multimodal EEG-language model. Self-supervised pretraining with natural language supervision enables label-efficient clinical phenotyping via a multiple-instance learning extension of the infoNCE loss.

Self-supervised deep learning for encoding pathological between-subject information in EEG data [link]

ICLR 2025, LMRL Workshop, Gijsen, S, Ritter, K

Active inference and the two-step task [link]

Scientific Reports 2022, Gijsen, S, Grundei, M, Blankenburg, F

Neural surprise in somatosensory Bayesian learning [link]

PLOS Computational Biology 2021, Gijsen, S, ..., Blankenburg, F

Experience

Postdoctoral Researcher in Machine Learning — *Germany*

Tübingen AI Center, Hertie Institute for AI in Brain Health — *Jan. 2023 - PRESENT*

- Self-supervised/representation learning: developed contrastive methods for physiological time series (EEG), including multi-modal settings with natural language supervision. Analyzed failure modes of self-distillation for noisy, high-dimensional signals and proposed fixes.
- Foundation models: for biological time series along two lines - multimodal contrastive pretraining and stabilized self-distillation - with strong downstream transfer and robust to distribution shifts.
- Generative modeling: developed diffusion transformer methods to generate realistic time series under out-of-distribution conditioning, enabling in-silico experiments.
- Led a team to 7th / 1,183 in the 2025 NeurIPS EEG Competition with a multi-modal fusion architecture

Research Assistant — *London, UK*

King's College London, Department of Neuroimaging — *Jul. 2017 - Sep. 2018*

- Pharmaco-imaging and analysis (time series, ICA) leading to two publications
- Project coordination and communication with scientists, radiologists, pharmacological industry, and medical staff

Internship — *London, UK*

King's College London, Department of Neuroimaging — *Nov. 2016 - Jul. 2017*

- ICA for pharmaco-imaging timeseries

Research Assistant — *Maastricht, Netherlands*

Maastricht University — *Aug. 2016 - Nov. 2016*

- Designing, programming, and piloting experimental work using high-field MRI

Education

PhD + Dr. rer. nat. in Computational Cognitive Neuroscience *Berlin, Germany*
Neurocomputation and Neuroimaging Unit, Freie Universität Berlin *Oct. 2018 - Dec. 2022*

- Computational modeling of learning and decision-making via probabilistic inference and Bayesian model comparison
- Competitive doctoral program (Mind and Brain - 10% Acceptance)
- Winner of DAAD International Research Scholarship (2 per year)
- Thesis: The brain as a generative model: information-theoretic surprise in learning and action [link]

MSc Research Master in Cognitive and Clinical Neuroscience *Maastricht, Netherlands*
Maastricht University *Oct. 2015 - Sep. 2017*

BSc Psychology *Maastricht, Netherlands*
Maastricht University *Oct. 2012 - Sep. 2015*

Selected Talks

[Poster] International Conference on Learning Representations (ICLR) *Rio de Janeiro, Brazil, 2026*

[Oral] Recent Advances in Machine Learning for Healthcare, PariSanté Campus *Paris, France, 2026*

[Poster] International Conference on Machine Learning (ICML) *Vancouver, Canada, 2025*

[Poster] LMRL Workshop @ International Conference on Learning Representations (ICLR) *Singapore, 2025*

[Oral] Neural Traces: Advanced M/EEG Methods and Clinical Applications *Berlin, Germany, 2024*

Skills

Techniques	Self-Supervised Learning, Representation Learning, Time series, Multimodal ML, Flow Matching, Hypothesis Testing
Languages	Python, SQL, MATLAB
Machine Learning	Pytorch (distributed training; DDP), Scikit-Learn, SciPy, PyMC, NumPy, Pandas
Tools/Infra	SLURM, Docker, Cloud (AWS/c-gen), Git, Bash
Communication	English (Fluent), Dutch (Mother Tongue), German (Intermediate)